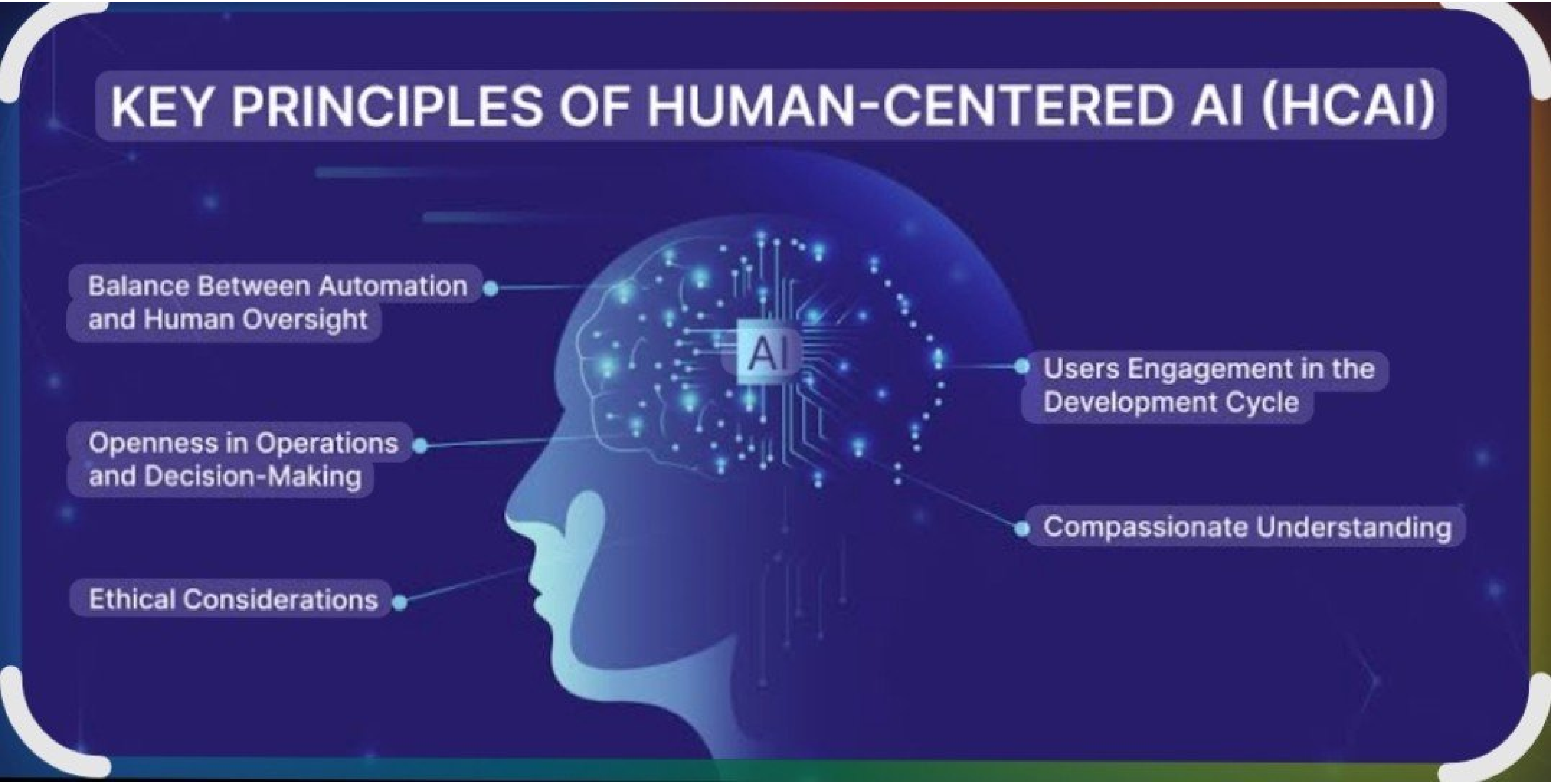
KEY PRINCIPLES OF HUMAN-CENTERED AI (HCAI)
Balance Between Automation and Human Oversight
AI
Users Engagement in the Development Cycle
Openness in Operations and Decision-Making
Compassionate Understanding
Ethical Considerations

Human-Centered Design Principles

1	User Needs First	Focus on real user requirements
2	Empathy	Understand and feel user perspectives
3	Personas & Testing	Build for groups, validate with feedback
4	Iterative Prototyping	Refine designs through ongoing testing
5	Relevant Visuals	Align aesthetics with target audience
6	Expert Collaboration	Involve specialists for deeper insights

1

Set clear and honest expectations

AI tools must clearly communicate their capabilities and limitations from the start to avoid users' misuse, confusion, over-reliance on the system, or deception. Setting accurate expectations helps prevent risky interactions while supporting safe, informed engagement.

2

Ensure human oversight and control

Balance AI automation and human agency by always allowing human operators to oversee, guide, control, and intervene in AI's operations. AI systems must provide accurate contextual cues and relevant information to support timely and effective human decision-making. People must always remain in control, especially in critical scenarios.

3

Design cognitively intuitive + physically accessible interfaces

The design of AI interfaces must account for humans' diverse physical, cognitive, and sensory capabilities, as well as their mental models, to support intuitive and accessible interactions that align with user expectations. As people from varied backgrounds and technical levels engage with these tools, context-aware interfaces will be key.

4

Provide system transparency

Give users transparency on the data the AI algorithm is built upon. Explain how this data is collected, curated, treated, used and stored by the system's algorithms. These inner workings on how the system operates should remain open to exploration and inspection.

5

Aid comprehension of AI's underlying systems

As AI tools and systems become more widespread, it's important to provide ways for users to gain a clear understanding of how the underlying algorithms work—how they make decisions, trigger actions, and lead to specific outcomes.

6

Work towards bias prevention and correction

Prevent the introduction of human bias and discriminatory practices in AI's data sets and algorithms. This is important to mitigate the occurrence of unfair treatment that perpetuates or amplifies social inequities. It is equally important to provide bias detection and reporting mechanisms for AI systems already in place.

7

Allow actions traceability and recovery from the unexpected

When errors or unexpected outcomes occur, the system must help the user understand what happened and why it happened, as well as provide options to reverse the action or change paths. The system must allow operators to trace back why this happens to determine the accountability of the system vs. human operators and inform future improvements.

8

Safeguard security and carefully manage risks

AI systems must anticipate and prevent risks arising from both human and system behaviors. This includes ensuring ethical, responsible, and secure interactions that protect personal privacy, prevent misuse, and mitigate security threats. Additionally, fail-safe mechanisms must be in place to allow users to respond to breaches, errors, or unsafe outcomes.

9

Respect human values and vulnerabilities

AI systems must be designed to uphold and respect fundamental human values, such as privacy, dignity, autonomy, and fairness, while supporting genuine human needs. They must not tamper with or exploit the cognitive, psychological, and emotional vulnerabilities of the human mind to manipulate, coerce, or deceive users.

10

Center AI solutions on human thriving

AI tools should be actively created with the intent to uplift human life. The capabilities of AI must be harnessed to augment human capabilities, enhance and streamline essential processes, empower people to reach higher potentials, and support both individual well-being and collective thriving. The goal is not AI for the sake of AI, the goal is a better future for humanity.